A 20-ps Time-to-Digital Converter (TDC) Implemented in Field-Programmable Gate Array (FPGA) with Automatic Temperature Correction

Weibin Pan, Guanghua Gong, and Jianmin Li

***Abstract*—This paper presents an automatic temperature correction design for carry chain based time-to-digital converter (TDC) in field-programmable gate array (FPGA). The bin-by-bin calibrations under different temperatures are performed for both plain TDC and Wave Union TDC to characterize the influence of temperature variation on the delay time of carry chain. Accordingly, a simplified temperature correction scheme by using a dedicated correction channel to measure the coefficient and correct fine time result for all TDC channels is implemented and tested. This method shows only few picosecond errors for both simulation and measurement. With this correction approach, a TDC precision of 21 ps has been achieved in Cyclone II FPGA under a wide ambient temperature range from 0°C to 60°C. Several design key points are also described in this paper.**

***Index Terms*—Field-programmable gate array (FPGA), temperature correction, time-to-digital converter.**

I. Introduction

CARRY chain structure in a field-programmable gate array (FPGA) can be used as the delay chain in time-to-digital converter for its intrinsic time resolution of tens of picoseconds. Benefiting from carry chain structure, multi-channel time-to-digital converters (TDCs) with very high precision have been implemented in single chip [1]–[7]. However, the delay time of carry chain is very sensitive to power supply voltage and temperature, thus the measurement precision could significantly deteriorate due to voltage and temperature (VT) variation [4], [5], [7].

A low dropout regulator (LDO) is capable of providing low noise power for FPGA core, and makes the effect of power fluctuation negligible. However, the temperature variation of FPGA is very difficult to reduce in many applications, thus a temperature compensation or correction scheme is essential for high precision measurement applications. The most common procedure is to perform a bin-by-bin calibration and update the look-up table (LUT) with temperature. If the signals to be measured are statistically random and with sufficiently high event rate, then the signals themselves can be used for calibration and the LUT can be updated by swapping ping-pong memories without introducing any dead time [2]; otherwise, a dedicated calibration signal is required, and dead time will be introduced when updating LUT since calibration signal and real hits cannot be injected to delay chain simultaneously.

This paper describes an automatic temperature correction scheme that does not depend on measured signals nor introduce any dead time. The bin-by-bin calibration is required only once after power up and a dedicated carry chain is instantiated to calibrate the variation of temperature and hence correct TDC result. The procedures can be fully performed on chip automatically.

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II. Methodology

The structure of carry chain based time-to-digital converter is shown in Fig. 1 [8]. An N-bit adder is instantiated and the sum is latched by a chain of registers. When a 0-to-1 transition occurs to the input signal, the carry signal of adder starts propagating and the sum of adder is latched by the TDC clock as a thermometer code. A thermometer-to-binary encoder is implemented to get the number of bin in which the carry signal locates. Since the delay times of cells (also called bin widths) throughout a carry chain are not identical, they should be calibrated, for example with the code density test [9]. An LUT is built and the content addressed by the bin number is the integrated delay time of all the previous bins. The output of LUT should be corrected by a temperature coefficient, which will be discussed in the later part of this paper. The obtained fine time result is then combined with the coarse time result. TDC channel skew caused by input signal routing delays in printed circuit board and die should be corrected.

A method called Wave Union [2], [3] can be implemented to introduce multiple 0-to-1 transitions to the delay chain. Hence multiple measurements can be performed and TDC precision will be improved by averaging multiple measurement results. In our design, a 2-transition Wave Union A is implemented and two transitions can be found simultaneously in the delay chain when a hit arrives. Wave Union TDC records these 2 transitions for encoder while plain TDC records only one of them. The sum of the bin numbers where the two transitions locate is treated as the effective bin number of Wave Union TDC.

A. Temperature Characteristic of Carry Chain

Cyclone II FPGA is based on 90-nm COMS technology and the propagation delay of a circuit is affected by the variations of

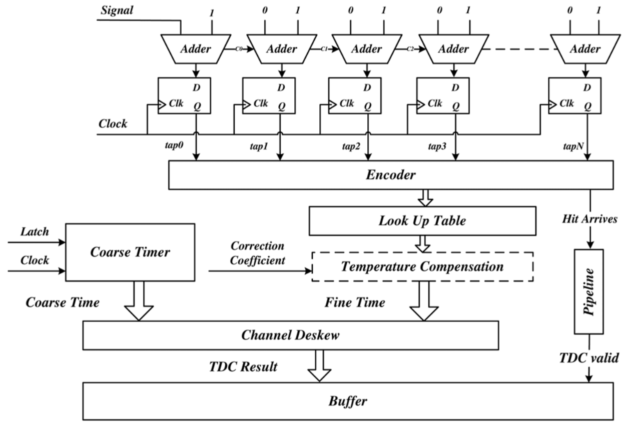

Fig. 1. Block diagram of carry chain based TDC.

carrier mobility and threshold voltage as the temperature fluctuates. Take COMS inverter as example, the propagation delay for the low-to-high transition can be derived [10]:

$$t_{pHL} = 0.69 R_{eqn} C_L. \tag{1}$$

Here R_{eqn} is the average "on" resistance of the NMOS transistor and C_L is the load capacitance:

$$R_{eqn} = \frac{3}{4}\frac{V_{DD}}{I_{DSAT}}\left(1 - \frac{7}{9}\lambda V_{DD}\right) \tag{2}$$

$$\text{with } I_{DSAT} = \mu_n C_{ox}\frac{W}{L}\left((V_{DD} - V_T)\, V_{DSAT} - \frac{V_{DSAT}^2}{2}\right). \tag{3}$$

Here λ, C_{ox} and ratio (W/L) are transistor parameters related to the manufacturing process, and may differ from different transistors. Supply voltage (V_{DD}) and drain-to-source voltage (V_{DSAT}) is fixed. Carrier mobility (μ_n) and threshold voltage (V_T) decreases with temperature and may be approximated by [11]

$$\mu(T) = \mu(T_r)\left(\frac{T}{T_r}\right)^{-k_\mu} \tag{4}$$

$$V_T(T) = V_T(T_r) - k_{vt}(T - T_r) \tag{5}$$

where T is the absolute temperature, T_r is room temperature, k_μ and k_{vt} are fitting parameters.

The intrinsic carrier mobility and threshold voltage of NMOS (PMOS) transistors in a die area can be considered approximately identical. Therefore the relative temperature dependence of each inverter is expected to be the same.

However, carry circuit is much more complicated than inverter, and the propagation time is difficult to be modeled accurately. Bin-by-bin calibrations for different temperatures would be much more effective to analyze the real temperature characteristic. Bin widths of the plain TDC under different temperatures are calibrated and shown in Fig. 2(a). The total delay time of valid bins is equal to the period of clock driving the carry chain, which in our design is 2.5 ns. Since Wave Union is an averaging procedure, the delay chain with Wave Union would have the same temperature characteristic as the raw delay chain.

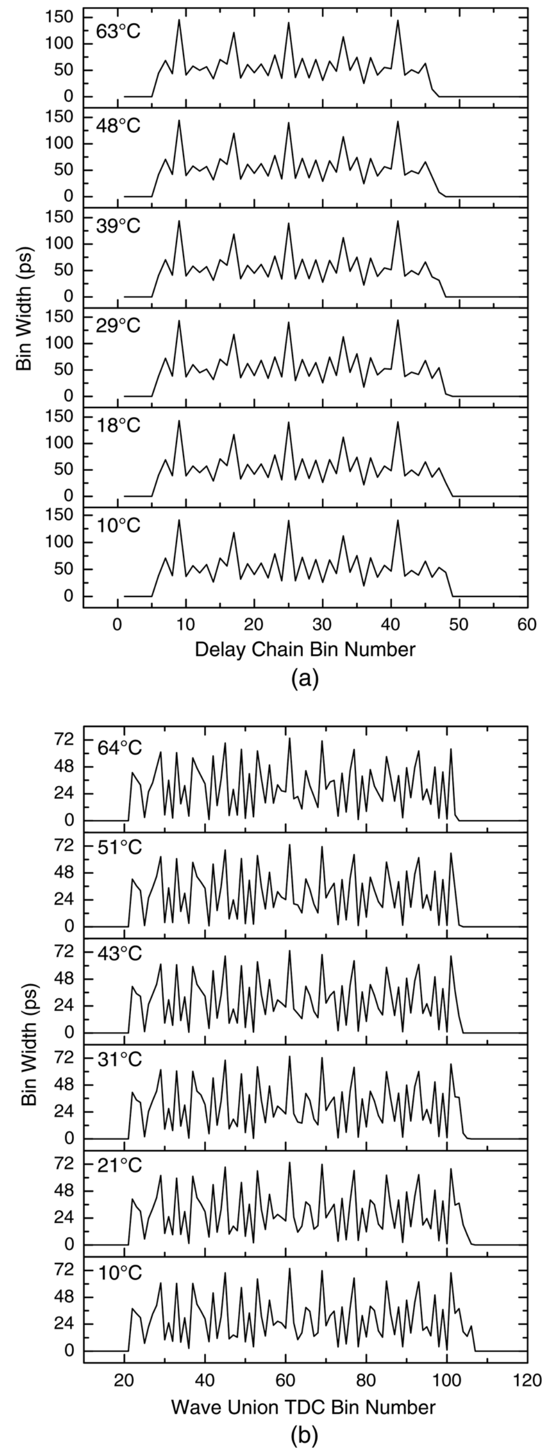

Fig. 2. Delay chain bin width under different temperatures, where (a) shows the bin width distribution of plain TDC, bins are valid from 15 to 50; (b) shows the bin width distribution of 2-transition Wave Union TDC, bins are valid from 20 to 110.

The bin widths of Wave Union delay chain under different temperatures are also calibrated and shown in Fig. 2(b).

The variability of manufacturing parameters result in the width differences of most bins. Besides, the ultra-wide bins in Fig. 2(a) locate in the boundary of Logic Array Block (LAB), and the higher C_L due to longer interconnect length leads to larger bin width. However, these factors are independent

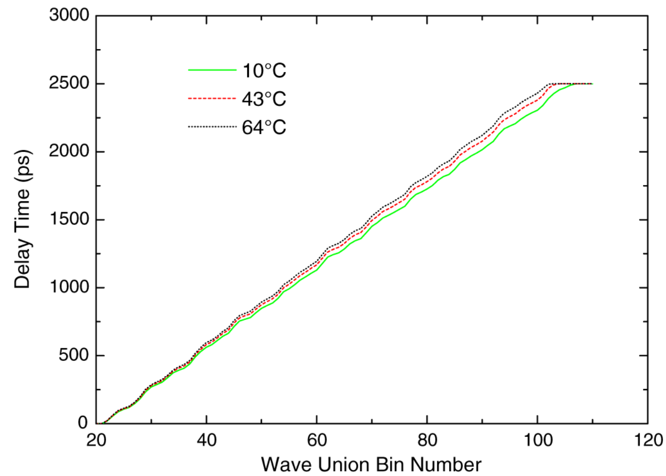

Fig. 3. Look-up tables (LUTs) under different temperatures.

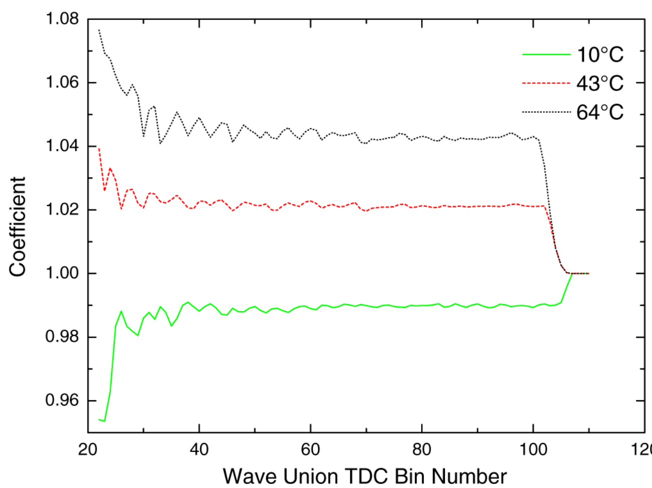

Fig. 4. LUT-temperature coefficients under different temperatures.

of temperature and make bin width distributions of different temperatures very similar, as shown in Fig. 2. We can assume that the whole delay chain expands or shrinks with variation of temperature but each bin contributes the same ratio to the whole delay chain. To verify this assumption, LUTs under different temperatures are calibrated, the differences are analyzed and shown in Fig. 3.

The normalized ratios of the LUTs to the one under room temperature (22°C), which we call temperature-LUT coefficients, are shown in Fig. 4.

Temperature-LUT coefficient of a certain temperature is very steady throughout the whole delay chain. Take coefficient of 10°C as example, the delay chain can be divided into three segments. For bins from 25 to 105, the coefficient fluctuation will cause a maximum time offset of 3.5 ps (at bin 102). For bins before 25, the integrated delay time is less than 120 ps, hence the relatively bigger coefficient deviation of 0.03 can only cause an insignificant time deviation of 3.6 ps. Bins after 105 are not used because they are out of one clock period (2500 ps). If the temperature-LUT coefficient is calibrated frequently, LUT

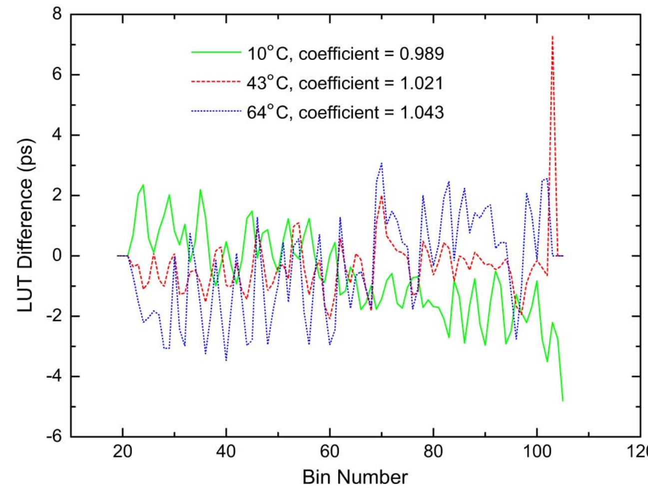

Fig. 5. Differences between the real calibrated and the deduced LUTs.

under any temperature can be deduced from the LUT of room temperature:

$$LUT_{temp} = LUT_{room} \times coefficient_{temp}. \tag{6}$$

Differences between the real calibrated and the deduced LUTs are illustrated in Fig. 5. LUT differences of most bins are in the range from -3 ps to 3 ps. Relative big differences can be seen at the end of delay chain, which is caused by incomplete width calibration for the last bin.

B. Temperature-LUT Coefficient Calibration

If the overall propagation time of the delay chain is longer than the period of driving clock, the logic transition may be recorded twice in the delay chain by two consecutive clock cycles [2], which are called *tap* and *tap_d*. Clock period drift caused by temperature is negligible and the jitter of clock has a much less impact than temperature variations. So the time difference between LUT(*tap*) and LUT(*tap_d*) is equal to the period of clock driving delay chain. Take the room temperature as example, we get the following equation:

$$LUT_{room}(tap) - LUT_{room}(tap\_d) = T0. \tag{7}$$

Similarly, if a transition is recorded twice at tap_d^* and tap^* under an unknown temperature, from (6) we can get

$$(LUT_{room}(tap^*) - LUT_{room}(tap\_d^*)) \times coefficient_{temp} = T0 \tag{8}$$

Then

$$coefficient_{temp} = \frac{LUT_{room}(tap) - LUT_{room}(tap\_d)}{LUT_{room}(tap^*) - LUT_{room}(tap\_d^*)}. \tag{9}$$

The precision of T0 calculated by $LUT_{room}(tap) - LUT_{room}(tap\_d)$ is limited by the TDC precision. Suppose that the TDC precision is 25 ps RMS, then T0 would have a precision of 35 ps (25 ps $\times \sqrt{2}$) RMS. By averaging N (65 536 in our design) samples of T0, we can get a much more precise value with a precision of $\frac{35\ ps}{\sqrt{N}}$ RMS. The averaged values of

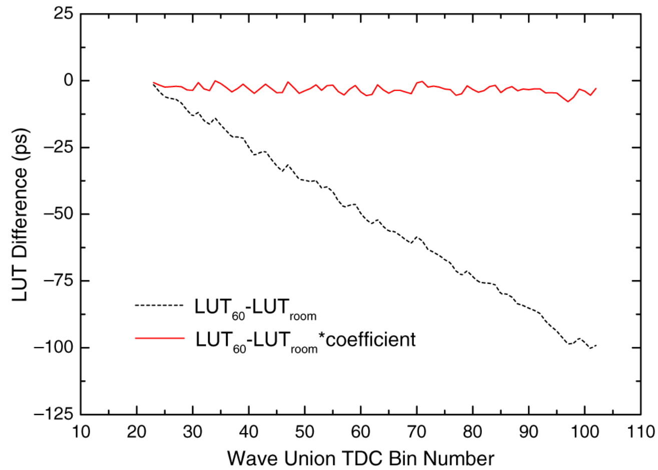

Fig. 6. Difference between LUT of 60°C and corrected/uncorrected LUT of room temperature.

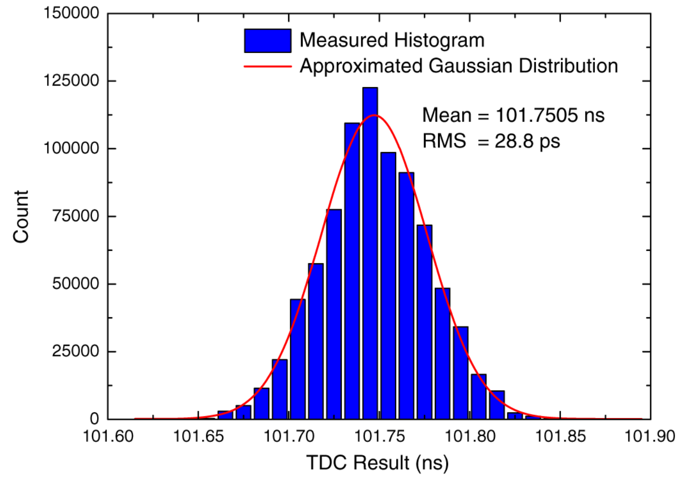

Fig. 8. Statistic histogram of 101.75-ns period measurement under temperature range of 0°C to 60°C.

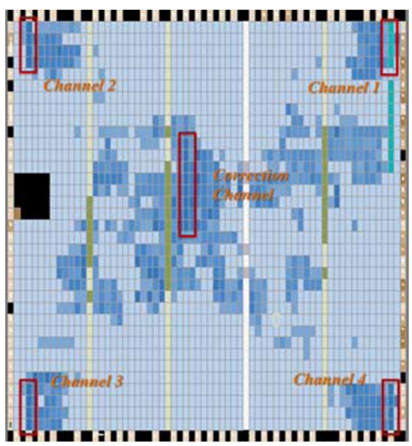

Fig. 7. Chip layout for one correction channel and four TDC channels.

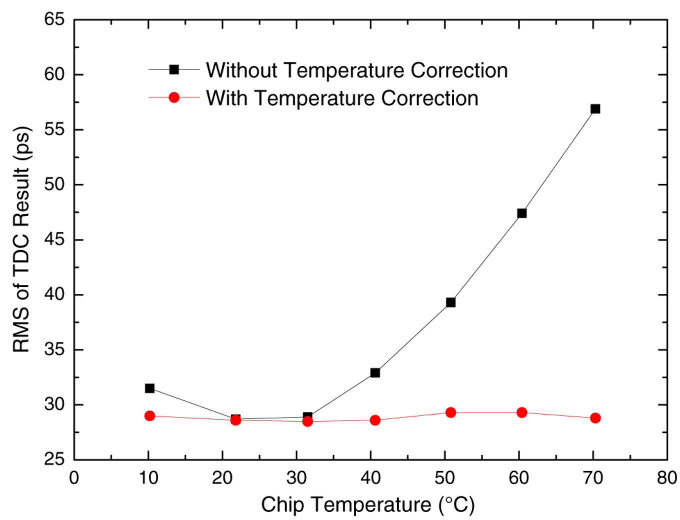

Fig. 9. TDC precision under different temperatures with and without temperature correction. The bin-by-bin calibration is taken under the chip temperature of 31°C.

T0 under room temperature and an unknown temperature are named T_{room} and T_{temp} respectively as in

$$T_{room} = \frac{\sum_{i=0}^{N-1} LUT_{room}(tap[i]) - LUT_{room}(tap\_d[i])}{N} \tag{10}$$

$$T_{temp} = \frac{\sum_{i=0}^{N-1} LUT_{room}(tap^*[i]) - LUT_{room}(tap\_d^*[i])}{N} \tag{11}$$

where *tap[i]* and *tap_d[i]* are the ith pairs of bins recorded in two consecutive clock cycles under room temperature, $tap^*[i]$ and $tap\_d^*[i]$ are the ith pairs of bins recorded under the unknown temperature.

The coefficient is calculated by dividing T_{room} by T_{temp} according to (9). Since temperature variation is a very slow process, the division function is designed with standard long division algorithm for its simple implementation and low resource utilization. In order to avoid decimals calculation, the dividend is multiplied by 2^{m}. In our design, variable m is equal to 8, which is the bit width of fine time result:

$$2^m \times coefficient_{temp} = \frac{2^m \times T_{room}}{T_{temp}}. \tag{12}$$

C. Real-Time Correction

In our design, an on-board 11.0592-MHz oscillator is used to generate calibration hits. There are 432 different phase relations between calibration signal and 400-MHz TDC clocks. The phase shift step is 5.787 ps, which is much smaller than the average bin width and guarantees calibration precision [12].

After power up, the calibration signal is fed into both the TDC channels and a dedicated correction channel. When bin-by-bin calibrations for all channels have been finished, TDC channels switch to measure the real physical hit and the correction channel starts temperature calibration. According to equation (12), the correction channel keeps updating coefficient by measuring the period of the driving clock of the TDC. Temperature corrected result is obtained by multiplying the raw fine time with the latest coefficient:

$$TDC_{fine} = \frac{LUT_{room}(bin) \times (2^m \times coefficient_{temp})}{2^m}. \tag{13}$$

Thanks to the multiplier resource in FPGA, the correction procedure can be done within one clock cycle. Therefore, the whole real time temperature correction process is performed in one chip.

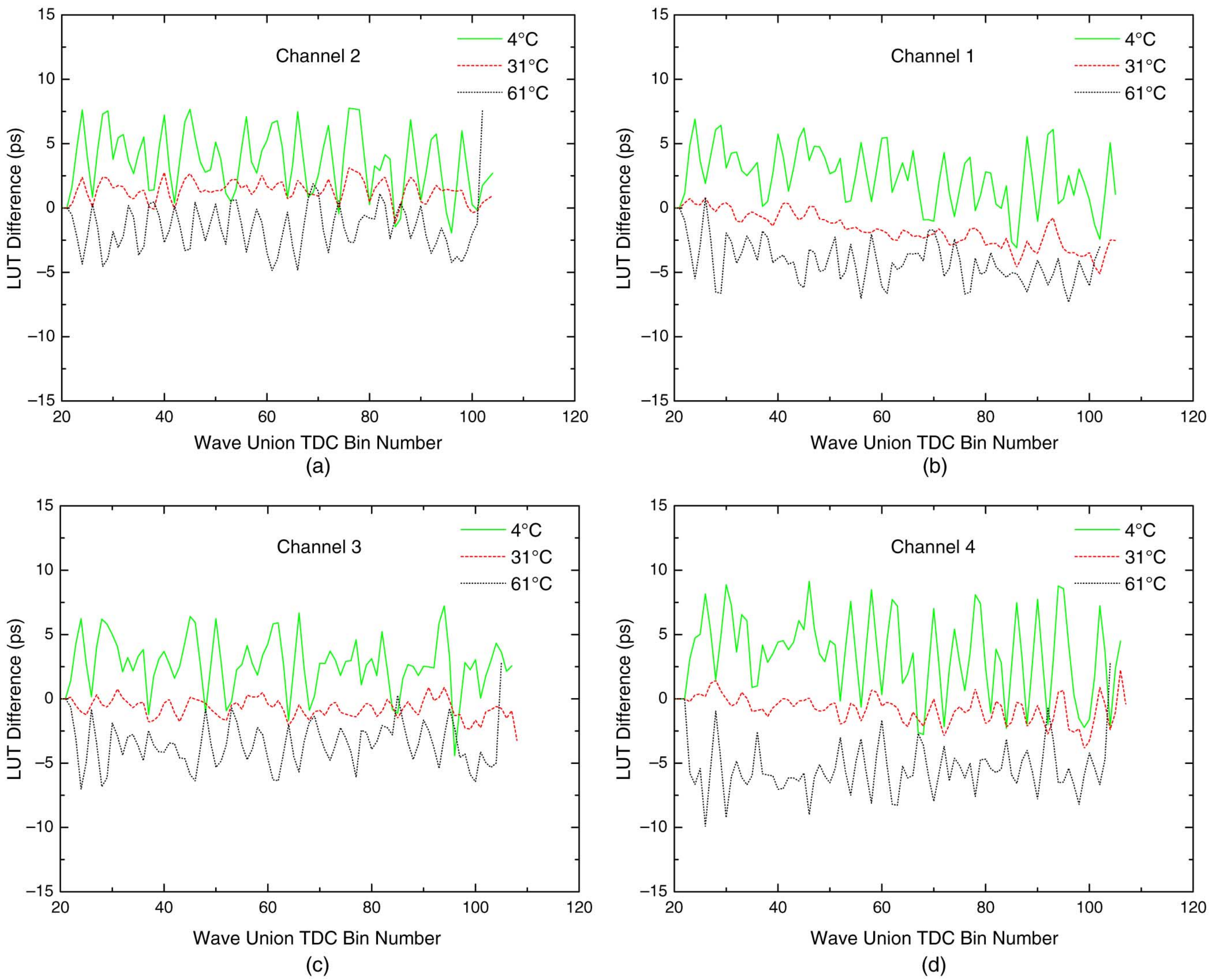

Fig. 10. Differences between the real calibrated LUTs and LUTs deduced from temperature-LUT coefficient. The temperature-LUT coefficient is calibrated in the correction channel and applied to four TDC channels. The bin-by-bin calibration is taken under the chip temperature of 30°C. (a) Wave Union TDC Bin Number. (b) Wave Union TDC Bin Number. (c) Wave Union TDC Bin Number. (d) Wave Union TDC Bin Number.

III. MEASUREMENTS

A. LUT Correctness

To evaluate the accuracy of coefficient correction, we calibrate the bin widths of delay chain at room temperature ($\sim 22^\circ\mathrm{C}$) and $60^\circ\mathrm{C}$ separately. The raw LUTs under these two temperatures (LUT_{room} and LUT_{60}) have a significant difference, as shown with a dash line in Fig. 6. By applying a $60^\circ\mathrm{C}$ coefficient correction to the LUT_{room}, the difference between the corrected LUT_{room} and LUT_{60} is shown as a solid line in Fig. 6, which has a mean value of 3 ps and a max value of 8 ps.

B. TDC Measurement

Four carry chains are placed at four corners of the FPGA chip while the dedicated correction carry chain is placed in the center. The chip layout is shown in Fig. 7.

A periodic test signal from a generator is injected into four TDC channels. The period of this test signal is set to 101.75 ns, which is able to fully evaluate the performance of fine time measurement while coarse time granularity is 2.5 ns.

A thermostatic container is used to control the environment temperature from $0^\circ\mathrm{C}$ to $60^\circ\mathrm{C}$ with a step of $10^\circ\mathrm{C}$. A temperature sensor is mounted on the bottom of FPGA to monitor the actual chip temperature. The bin-by-bin calibration is taken under the chip temperature of $31^\circ\mathrm{C}$.

More than 100 000 TDC results are recorded under each temperature. The statistic histogram of TDC channel 1 under the whole temperature range is depicted in Fig. 8. TDC results of four channels are listed in Table I. The precisions are not identical due to different bin width distributions, and the worst precision of single shot measurement is 21.8 ps ($30.9\ \mathrm{ps}/\sqrt{2}$).

According to Fig. 9, the temperature variation has no influence on the precision of the TDC with temperature correction.

TABLE I
TDC RESULTS OF FOUR CHANNELS

TDC Channel	Mean	RMS
1	101.7505 ns	28.8 ps
2	101.7505 ns	30.9 ps
3	101.7505 ns	29.5 ps
4	101.7505 ns	30.1 ps

TABLE II
LOGIC UTILIZATION OF TEMPERATURE CORRECTION

TDC	Resource	Used	Available	Utilization
No Temperature Correction	Combinational functions	21,957	33,216	66%
	Logic Registers	25,868	33,216	78%
	Memory bits	211,282	483,840	44%
	Multiplier 9-bit	0	70	0%
With Temperature Correction	Combinational functions	23,494	33,216	71%
	Logic Registers	28,085	33,216	85%
	Memory bits	213,586	483,840	44%
	Multiplier 9-bit	64	70	91%

IV. DISCUSSION

A. Correction Channel

Temperature calibration can be performed in each TDC channel or just in one dedicated correction channel. With the first approach, the real physical hit is used to calibrate temperature and an individual temperature-LUT coefficient is obtained for each TDC channel. However, each channel requires a longer carry chain that takes more resources. Moreover, in order to get a precise coefficient, the real physical hit should be sufficiently frequent and random that could be recorded in different bins.

With the second approach, the dedicated calibration hits generated by an oscillator is used for both bin-by-bin calibration and temperature calibration. The coefficient calculated from the correction channel is shared for all TDC channels. Thus the coefficient consistency between different TDC channels must be verified.

The bin widths of four channels and coefficient under different temperatures are calibrated. Differences between the real calibrated LUTs and LUTs deduced from temperature-LUT coefficient are calculated and illustrated in Fig. 10. The ultra large difference at the end of delay chain is due to incomplete calibration for the last bin. Except for that, the maximum deviation from the real LUT value is 10 ps, which is smaller than the average bin width.

B. Logic Utilization of Temperature Correction

The entire design including 32-channel TDC and Ethernet readout logic is implemented in an Altera EP2C35F484C6 device. The logic utilizations with and without temperature correction are compared and shown in Table II. The temperature correction logic takes 5% of combinational functions, 7% of logic registers, and 64 multiplier 9-bit elements (2 per TDC channel).

V. CONCLUSION

An automatic temperature correction method for carry chain based TDC has been implemented in an Altera EP2C35F484C6 device. We present the analysis of temperature characteristic of plain TDC and Wave Union TDC, and the design detail of temperature calibration and real time correction. With the proposed temperature correction technique, the TDC can achieve a precision of 21.8 ps in a wide ambient temperature range from 0°C to 60°C and no precision deterioration is observed.

The advantages of this temperature correction approach include its independency of measured signals, no requirement for external component (e.g., temperature sensor), no introduction of dead time and high integration.

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